

Project: Causal Machine Learning

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1 Introduction

We consider estimation of the effect of eligibility and participation in 401(k) (this is a tax-advantaged retirement savings plan) on accumulated assets using 9915 observations at the household level, drawn from the Survey of Income and Program Participation:

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library(hdm)
data(pension)
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In this project, each data analysis question must be based on debiased/targeted machine learning (unless otherwise stated). Make sure to use 5- or 10-fold cross-fitting in all such analyses, and to base these either on a library of complementary algorithms, or a single, but carefully tuned algorithm. In all data analyses, you are expected to report an estimate and 95% confidence interval, and to interpret the results in an accessible manner. In addition, you are expected to make some checks to evaluate and report the quality of the results (e.g., you may want to check the stability of inverse probability weights, or whether some individuals have extreme influence functions).

2 Task

The key problem in determining the effect of participation in 401(k) plans on accumulated assets is the fact that the decision to enroll in a 401(k) is non-random. It is generally recognized that some people have a higher preference for saving than others. It also seems likely that those individuals with high unobserved preference for saving would be most likely to choose to participate in tax-advantaged retirement savings plans and would tend to have otherwise high amounts of accumulated assets. The presence of unobserved savings preferences with these properties then implies that conventional estimates that do not account for confounding will be biased upward, tending to overstate the savings effects of 401(k) participation.

We will first focus on **the effect of eligibility for enrolling in a 401(k) plan**. Around the time 401(k)'s initially became available, people were unlikely to be basing their employment decisions on whether an employer offered a 401(k) but would instead focus on income. Thus, eligibility for a 401(k) could be taken as exchangeable conditional on income, and the causal effect of 401(k) eligibility could be directly estimated by appropriate comparison across eligible and ineligible individuals with the same income.

1. Explain why the above reasoning makes it plausible that eligibility for a 401(k) is exchangeable conditional on income. You may consider the use of a causal diagram to support your reasoning.
2. Use AIPW **as well as** TMLE to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets, controlling for age, income, family size, years of

education, a married indicator, a two-earner status indicator, a defined benefit pension status indicator, an IRA (individual retirement account) participation indicator, and a home ownership indicator. Report either the ATE or the ATT (or both), depending on what seems most suitable to you. Compare with the corresponding estimate when no covariate adjustment is made.

3. The previous analysis can be used to test if eligibility for enrolling in a 401(k) plan, A , affects net financial assets Y . A more powerful test is typically obtained by testing whether $E\{\text{Cov}(A, Y|L)\} = 0$, with L the control variables listed in question 2. The efficient influence curve of $\theta = E\{\text{Cov}(A, Y|L)\}$ is

$$\{A - E(A|L)\} \{Y - E(Y|L)\} - \theta.$$

Use this to estimate θ and its standard error, and report a p-value of the test of the null hypothesis that θ equals 0.

4. Use the R-learner to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets in function of age, income and years of education, but controlling for the variables listed in question 2. Report a histogram of the effect estimates obtained for all individuals in the study. In addition, visualize¹ how the effects vary by age, income and years of education.
5. This is a more difficult question. The counterfactual net financial assets that would be observed when being eligible for enrolling in a 401(k) plan can be predicted in function of age, income and years of education (controlling for the variables listed in question 2) using the same strategy as explained for DR-learner, but using pseudo-outcome

$$C_i = \hat{Y}_i^1 + \frac{A_i}{\hat{p}_i} (Y_i - \hat{Y}_i^1)$$

following the notation in the slides. Let us denote the predictions obtained via the DR-learner based on these pseudo-outcomes as $\tilde{Y}^1$. First, obtain such predictions and show them in a histogram. Next, for the obtained predictions, estimate the counterfactual prediction error

$$E \left\{ \left(Y^1 - \tilde{Y}^1 \right)^2 \right\},$$

along with a 95% confidence interval. Hint: This can be done by slightly adapting one of the causal machine learning strategies discussed in the course. You do not need to (be able to) derive efficient influence curves for being able to answer this question.

Make a **concise, but complete** report describing what analyses you performed and what are your conclusions to the various questions. There is no need to be extensive in your report. For instance, there is no need to repeat the theory that was used, as you may assume that the reader is familiar with the procedures that you used. However, you should carefully interpret the reported estimates and make sure that their meaning is clear and not obscure. Overall, I expect that 2-3 pages should suffice; 4 pages is a maximum.

¹It suffices to visualize marginal associations; that is, to run separate analyses for the conditional average treatment effect given only age, or only income, or only education.

You can work on this project in groups of 2 or 3 people. It is recommended that you split the work, **and then give feedback on each other's work and report**. If you split the work, the project should not take too much time, but otherwise it may. In choosing a way to split the work, note that question 1 does not require a lot of work (as it does not require analysis of the data). A logical split in 3 would be questions 1-3 for one person, question 4 for one person and question 5 for one person. One report per group must be returned no later than no later than May 16, together with structured software code (i.e. a script file in R or Python). All the files that you submit, must carry the family name of one of the members in the group.

Make sure your report briefly mentions who did what. One third of the score will be on your own contribution and two thirds on the project as a whole (as indeed you are also responsible for the entirety of the project).